SENIOR PROJECT · COMPUTER SCIENCE

Fine-Grained Hallucination Detection

& Severity-Aware Mitigation in Vision-Language Models

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Introduction

Strength

LVLMs (e.g., LLaVA) handle image-grounded QA with strong fluency.

Problem

They hallucinate visual content not supported by the image.

Error Types

Objects, attributes, relationships, counts, and scene details.

Severity Varies

From minor wording slips to serious factual mismatches.

Impact

Reduces trust in accessibility, medical, and robotics applications.

Motivation & Research Gap

THE PROBLEM

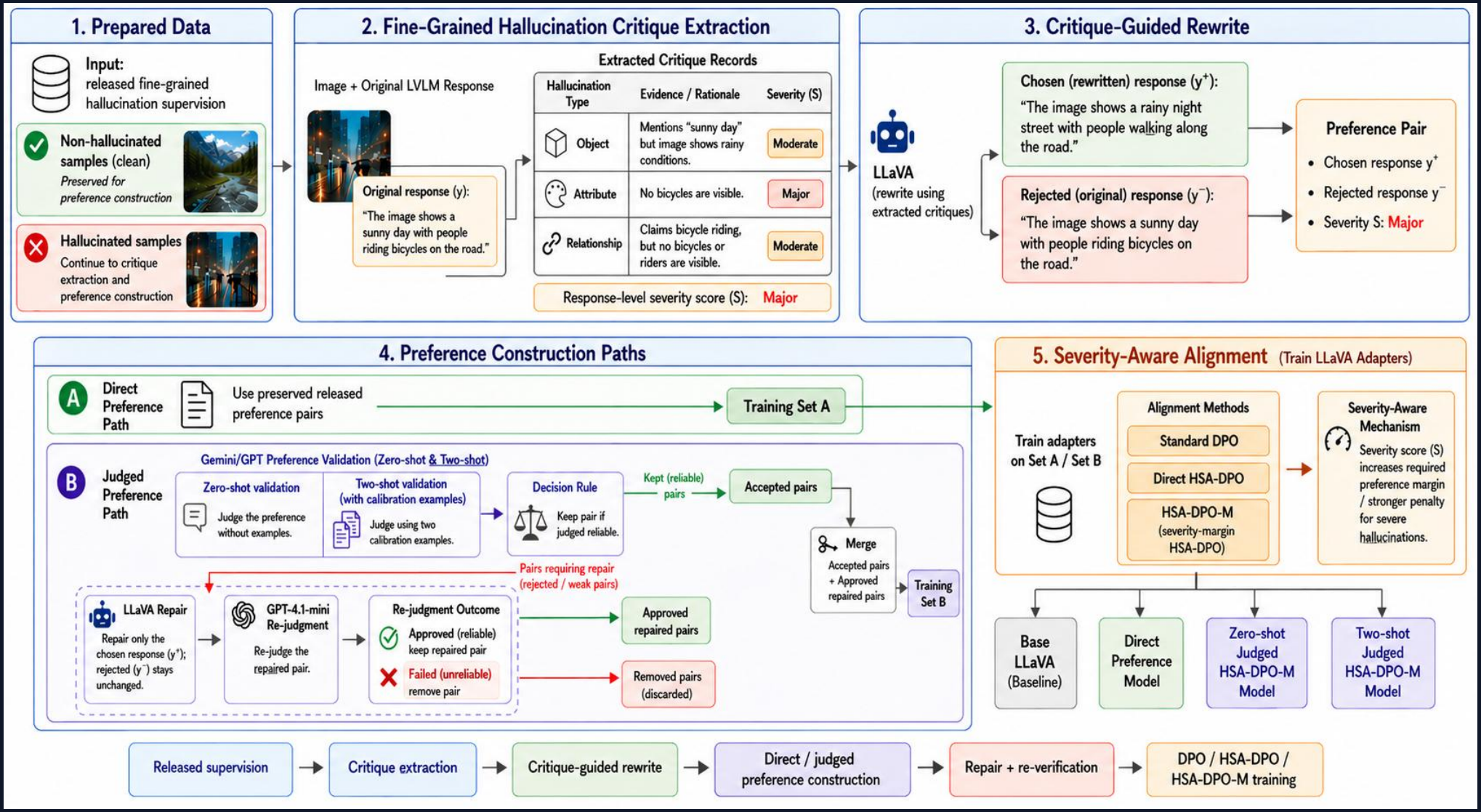
- Most prior work treats hallucination as a coarse **binary** problem.
- Binary labels miss **what** is wrong, **where**, and **how severe**.
- DPO alignment depends on reliable chosen / rejected pairs.
- Auto-generated pairs are noisy without external validation.

OUR APPROACH

One unified pipeline that combines:

- Fine-grained critiques (type + location)
- Severity scores for weighted training
- Judge-validated preference pairs
- Repair + re-verification of weak pairs
- Severity-margin DPO training

Pipeline Overview



Compared Preference Training Approaches

Variant	Preference Data	Severity Use
Standard DPO	Uses chosen/rejected response pairs directly	Does **not** use severity scores
Direct HSA-DPO	Uses released/prepared preference pairs directly	Uses severity to penalize severe hallucinations more strongly
Judged Severity-Margin HSA-DPO	Preference pairs are first checked by Gemini/GPT-style judges; weak pairs are repaired or removed	Uses severity as a margin, requiring stronger preference separation for severe hallucinations

Experiment Setup

MODELS

Base: LLaVA-1.5-7B

Variants: DPO, Direct HSA-DPO, HSA-DPO-M (zero-shot & two-shot)

Judges: Gemini-2.5-Flash-Lite, GPT-4o-mini · Repair: GPT-4.1-mini

DATA

16,143 normalized records

7,643 hallucinated + 8,500 clean

→ **8,386** preference pairs

TRAINING

1 epoch · effective BS 32 · LR 2e-6

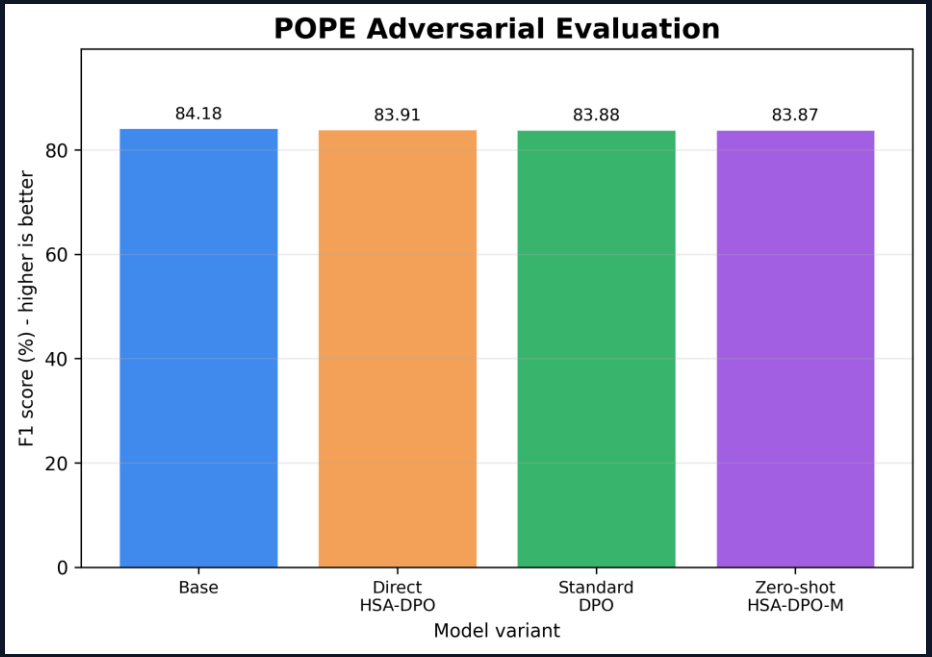
GPU: NVIDIA RTX 6000 Ada Generation, 48 GB VRAM (rented)

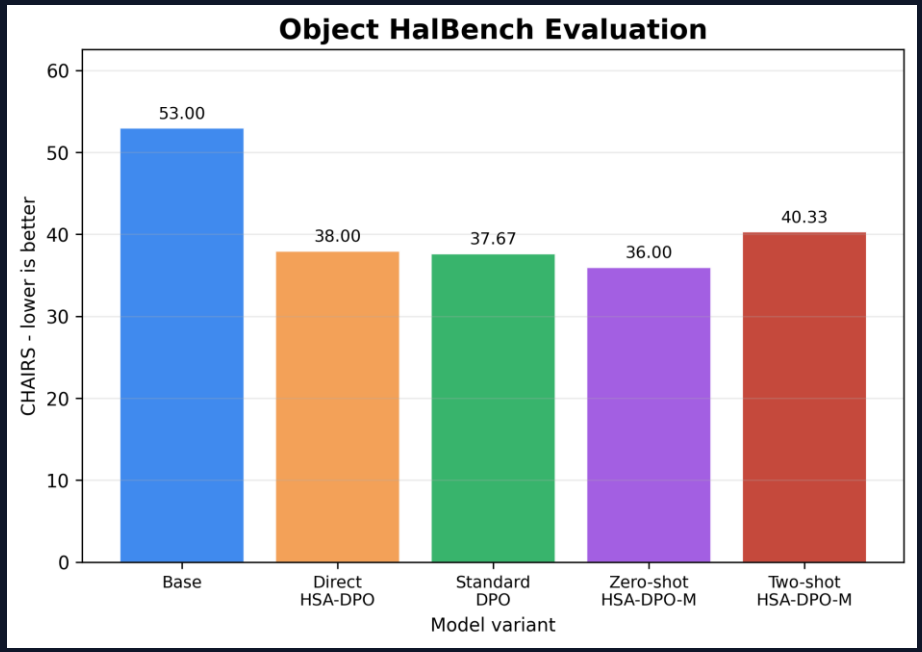
BENCHMARKS

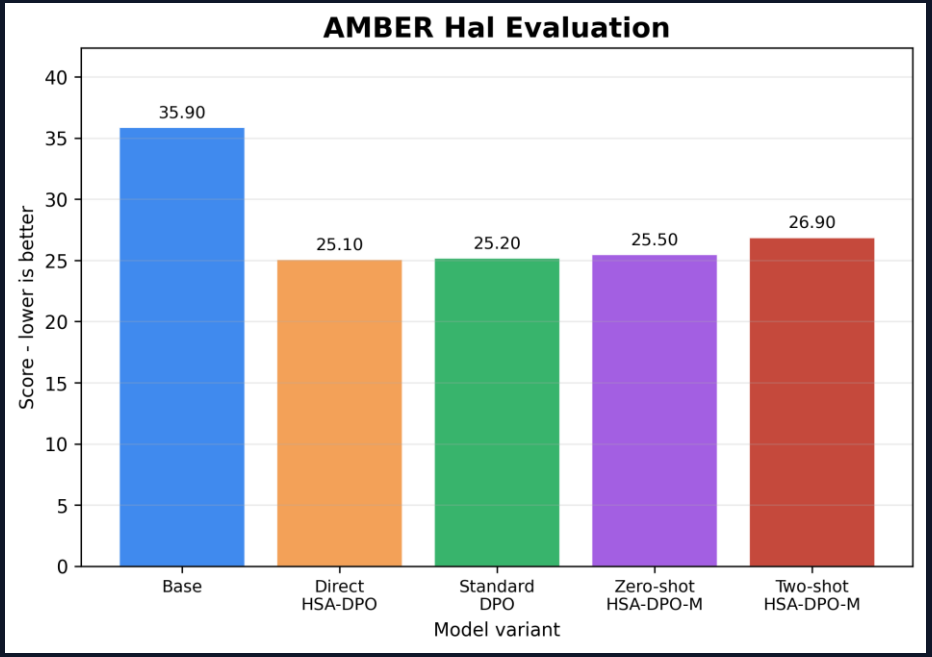
- **POPE** — yes/no object questions
- **Object HalBench** — CHAIRs / CHAIRi
- **AMBER** — CHAIR + Hal rate

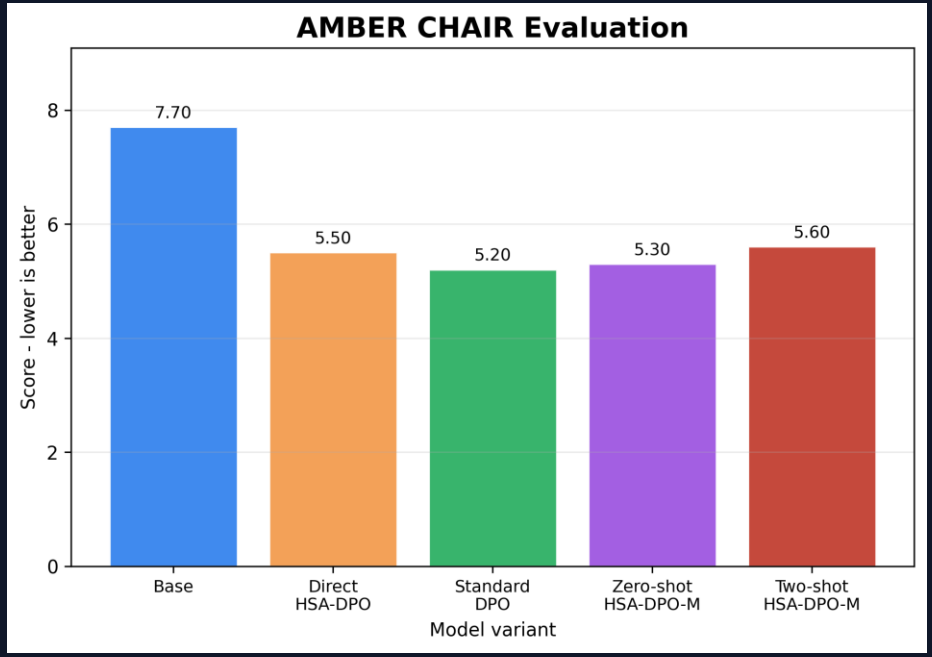
Benchmark Results Summary

Benchmark	Metric	Base LLaVA-1.5-7B	Best Mitigation Result	Change
POPE Adv.	F1 ↑	84.18	83.91	-0.27
POPE Adv.	Precision ↑	89.92	90.70	+0.78
POPE Adv.	Recall ↑	79.13	78.07	-1.06
Object HalBench	CHAIRs ↓	53.00	36.00	-17.00
Object HalBench	CHAIRi ↓	15.72	10.41	-5.31
AMBER	AMBER-CHAIR ↓	7.70	5.20	-2.50
AMBER	AMBER-Hal ↓	35.90	25.10	-10.80
AMBER	AMBER-Cog ↓	4.30	2.50	-1.80


POPE Adversarial Evaluation
F1 score (%) - higher is better
0
20
40
60
80
84.18
83.91
83.88
83.87
Base
Direct HSA-DPO
Standard DPO
Zero-shot HSA-DPO-M
Model variant


Object HalBench Evaluation
CHAIRS - lower is better
0
10
20
30
40
50
60
53.00
38.00
37.67
36.00
40.33
Base
Direct HSA-DPO
Standard DPO
Zero-shot HSA-DPO-M
Two-shot HSA-DPO-M
Model variant


AMBER Hal Evaluation
Score - lower is better
0
5
10
15
20
25
30
35
40
35.90
25.10
25.20
25.50
26.90
Base
Direct HSA-DPO
Standard DPO
Zero-shot HSA-DPO-M
Two-shot HSA-DPO-M
Model variant


AMBER CHAIR Evaluation
Score - lower is better
0
2
4
6
8
7.70
5.50
5.20
5.30
5.60
Base
Direct HSA-DPO
Standard DPO
Zero-shot HSA-DPO-M
Two-shot HSA-DPO-M
Model variant

Discussion & Conclusion

KEY FINDINGS

- Fine-grained critiques + severity scores enable type-aware mitigation.
- Judge-validated, repaired pairs improve training signal quality.
- HSA-DPO-M cuts **Object HalBench CHAIRs by 17 pts**.
- AMBER Hal drops by **10.8 pts** vs. base.

LIMITATIONS

- POPE F1 not improved over base; gains are not universal.
- Severity-aware training increases caution, slightly reducing recall.
- Two-shot variant underperforms zero-shot HSA-DPO-M.

FUTURE WORK

- Stronger preference filtering and judging strategies.
- Higher-quality repair models for rejected pairs.
- Broader benchmark coverage to test generalization.